200mA, 30V Dual Schottky Barrier Diode

FEATURES

- AEC-Q101 qualified
- Fast switching speed
- Low forward voltage
- Surface mound device type
- Moisture sensitivity level: level 1, per J-STD-020
- RoHS Compliant
- Halogen-free

KEY PARAMETERS		
PARAMETER	**VALUE**	**UNIT**
I_F	200	mA
V_{RRM}	30	V
I_{FSM}	600	mA
$T_{J\ MAX}$	125	°C
V_F at I_F=100mA	800	mV
Configuration	Dual die	

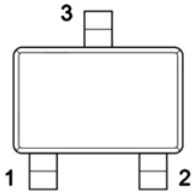
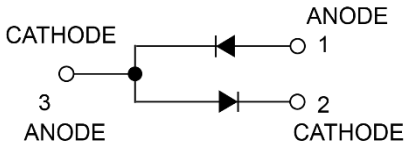
APPLICATIONS

- Ultra high-speed switching
- Line termination
- Voltage clamping
- Reverse polarity protection

Pb-Free RoHS COMPLIANT HALOGEN FREE

MECHANICAL DATA

- Case: SOT-323
- Molding compound meets UL 94V-0 flammability rating
- Terminal: Matte tin plated leads, solderable per J-STD-002
- Meet JESD 201 class 2 whisker test
- Weight: 5.00mg (approximately)

PACKAGE: SOT-323	PIN CONFIGURATION	CIRCUIT DIAGRAM
		

ABSOLUTE MAXIMUM RATINGS (T_A = 25°C unless otherwise noted)				
PARAMETER		**SYMBOL**	**VALUE**	**UNIT**
Power dissipation(1)		P_D	200	mW
Repetitive peak reverse voltage		V_{RRM}	30	V
Forward current		I_F	200	mA
Repetitive peak forward current		I_{FRM}	300	mA
Non-repetitive peak forward surge current	t < 1.0s	I_{FSM}	600	mA
Junction temperature		T_J	-55 to +125	°C
Storage temperature		T_{STG}	-55 to +150	°C

Note:

1. Device mounted on an FR4 Printed-Circuit Board (PCB), single-sided copper, tin-plated and standard footprint

THERMAL PERFORMANCE			
PARAMETER	**SYMBOL**	**TYP**	**UNIT**
Junction-to-ambient thermal resistance(1)	$R_{\Theta JA}$	500	°C/W

Thermal Performance Note:

1. Device mounted on an FR4 Printed-Circuit Board (PCB), single-sided copper, tin-plated and standard footprint

ELECTRICAL SPECIFICATIONS (T_A = 25°C unless otherwise noted)

PARAMETER	CONDITIONS	SYMBOL	MIN	TYP	MAX	UNIT
Forward voltage(1)	I_F = 0.1mA	V_F	-	-	240	mV
	I_F = 1mA		-	-	320	
	I_F = 10mA		-	-	400	
	I_F = 30mA		-	-	500	
	I_F = 100mA		-	-	800	
Reverse breakdown voltage	I_R = 100μA	V_{BR}	30	-	-	V
Reverse current(2)	V_R = 25V	I_R	-	-	2	μA
Junction capacitance	f = 1MHz, V_R = 1V	C_J	-	-	10	pF
Reverse recovery time	I_F = I_R = 10mA, I_{RR} = 1mA, R_L = 100Ω	t_{rr}	-	-	5	ns

Notes:

1. Pulse test with PW=0.3 ms
2. Pulse test with PW=30 ms

ORDERING INFORMATION

ORDERING CODE	PACKAGE	PACKING
BAT54SWH RFG	SOT-323	3,000 / 7” Tape & Reel

CHARACTERISTICS CURVES

(T_A = 25°C unless otherwise noted)

Fig.1 Power Dissipation Curve

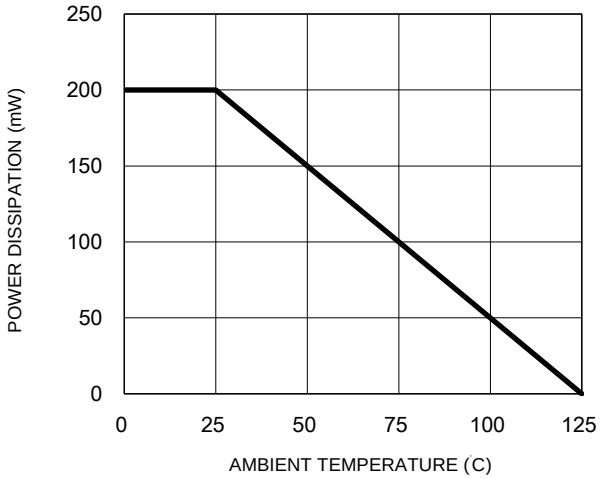


Fig.2 Typical Junction Capacitance

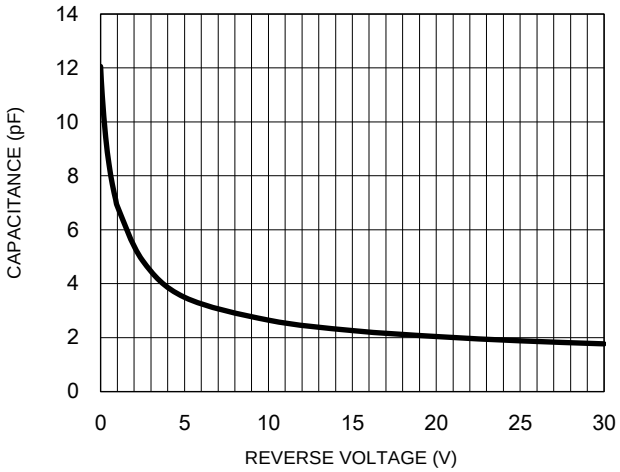


Fig.3 Typical Reverse Characteristics

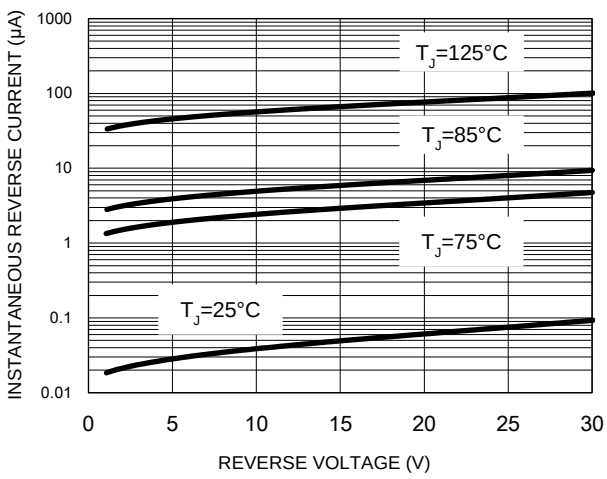
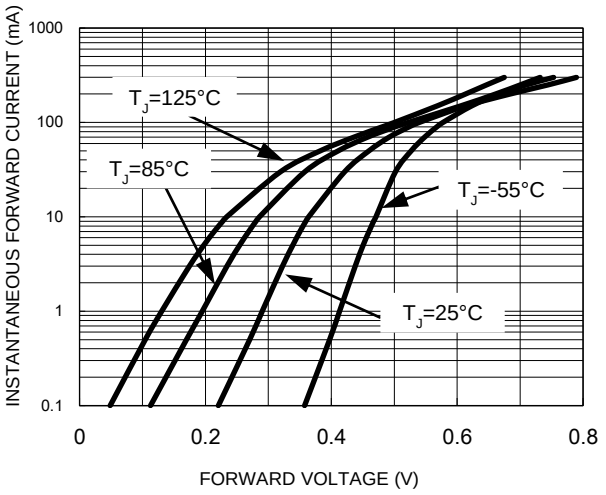
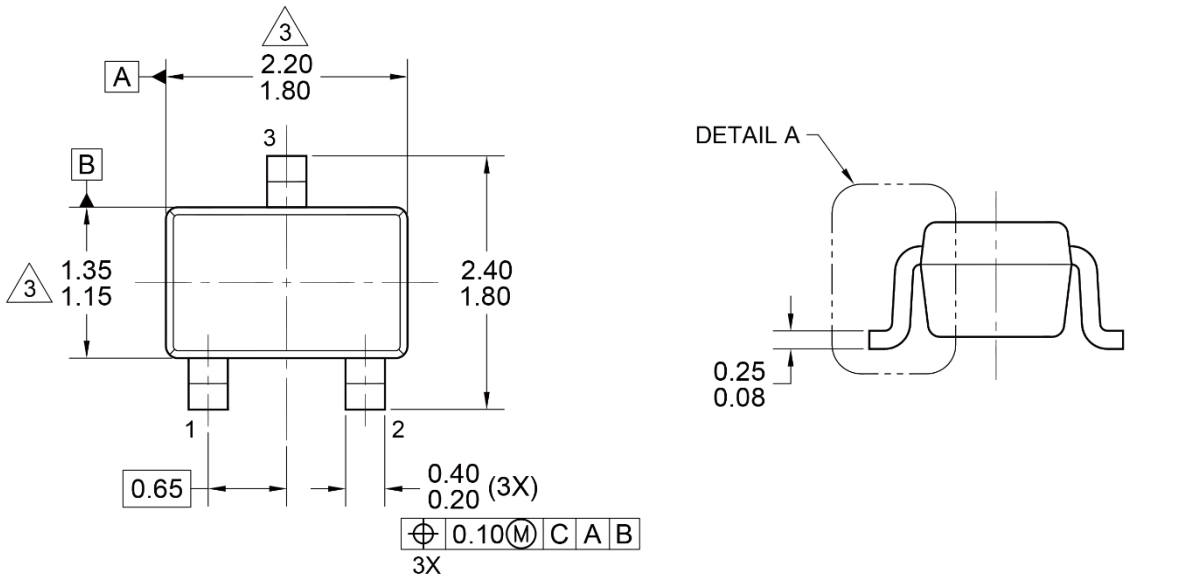


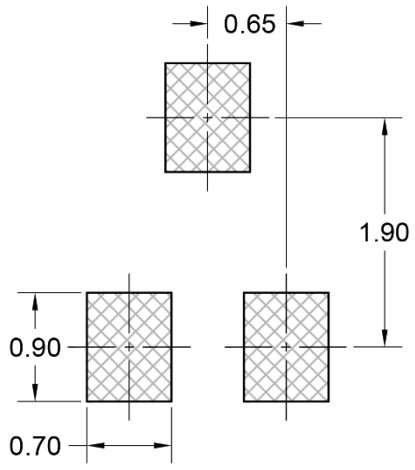
Fig.4 Typical Forward Characteristics



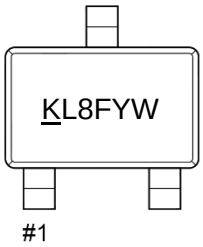
PACKAGE OUTLINE DIMENSIONS

SOT-323





SUGGESTED PAD LAYOUT



MARKING DIAGRAM

NOTES: UNLESS OTHERWISE SPECIFIED

1. ALL DIMENSIONS ARE IN MILLIMETERS.
2. DIMENSIONING AND TOLERANCING PER ASME Y14.5M-1994.
3. MOLDED PLASTIC BODY DIMENSIONS DO NOT INCLUDE MOLD FLASH, PROTRUSIONS OR GATE BURRS.
4. DWG NO. REF: HQ2SD07-SOT323-098 REV C.

KL8 = Device marking
F = Factory code
Y = Year code
W = Bi-Week code (A~Z)

Notice

Specifications of the products displayed herein are subject to change without notice. TSC or anyone on its behalf, assumes no responsibility or liability for any errors or inaccuracies.

Purchasers are solely responsible for the choice, selection, and use of TSC products and TSC assumes no liability for application assistance or the design of Purchasers' products.

Information contained herein is intended to provide a product description only. No license, express or implied, to any intellectual property rights is granted by this document. Except as provided in TSC's terms and conditions of sale for such products, TSC assumes no liability whatsoever, and disclaims any express or implied warranty, relating to sale and/or use of TSC products including liability or warranties relating to fitness for a particular purpose, merchantability, or infringement of any patent, copyright, or other intellectual property right.

The products shown herein are not designed for use in medical, life-saving, or life-sustaining applications. Customers using or selling these products for use in such applications do so at their own risk and agree to fully indemnify TSC for any damages resulting from such improper use or sale.